

Can international supply chain induce a return premium? Evidence from U.S. leading high-technology firms and Taiwan stock market

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Highlights

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This study examines significant return spillover effects from the U.S. top high-technology firms to their Taiwanese suppliers, electronic industry indices and market index in Taiwan.

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We find returns on these U.S. stocks prominently spillover to Taiwanese non-electronic index, possibly resulting from investor sentiment diffusion rather than fundamentals connections.

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Empirical results show that Taiwanese stocks with higher sensitivity to U.S. returns have stronger idiosyncratic volatility.

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Stocks with highest sensitivity to U.S. returns, have a return premium and highest abnormal returns.

Abstract

This study aims to explore relationships between return linkages on international supply chain, idiosyncratic volatility and abnormal returns in local markets. Empirically, focusing on the seven top U.S. high-technology firms and local market of their Taiwanese suppliers, we find significant daily return spillovers from U.S. stocks to the suppliers, electronic industry indices, market index, and non-electronic index in Taiwan, possibly resulting from

fundamentals connections or investor sentiments diffusion. Furthermore, stocks strongly sensitive to the U.S. stocks, have high idiosyncratic volatility and a return premium. Moreover, we provide an external sensitivity factor for asset pricing in Taiwan.

Introduction

Contradictory to the traditional asset pricing model (CAPM), Merton (1987) posits stocks with higher idiosyncratic volatility (henceforth IV) should have higher expected returns because investors, with limit access to information, cannot perfectly diversify the idiosyncratic risk. Empirically testing Merton (1987)'s theory in the U.S. stock market, Ang et al. (2006) find negative relationship between expected returns and IV. However, Boehme et al. (2009) examine that for those stocks with low visibility and short-sale constraints, the relationship becomes positive. Actually, since IV is usually defined as standard deviation of the residuals from the Fama and French (1993) model, it can be interpreted as a risk exposure of the missing factors (Chen and Petkova, 2012). Extensive literature focuses on explaining the pricing puzzle of IV or explores missing factors in views of investor preferences, investor irrationality, market characteristics such as arbitrage asymmetry (Stambaugh et al., 2015), or firm specific characteristics like investment and profitability (Malagon et al., 2015).

It's noted that previous studies on the IV mainly analyze risk factors from local stock market. However, in the context of international markets, not only information is limited, but also delivery of information could be distorted (Ahearne et al., 2004). One of the possible channels for information delivery or risk transmission across stock markets is the global supply chain. In the background of local market, Lauren and Andrea (2008) find that the prediction of one stock return by returns on its major customers can be attributed to investors' limited attention and under-reaction. Motivated by Lauren and Andrea (2008), we firstly hypothesize that international return spillovers exist along global supply chain. Furthermore, it's known that asset returns are driven by information. If the first hypothesis is verified, another hypothesis is put forward that return linkages or sensitivities to foreign stocks, possibly affected by distorted information, could be mispriced.

To test above two hypotheses, we focus on return spillovers between leading U.S. high-technology firms and local market of their Taiwanese suppliers. In January 2018, Thomson Reuters named the world's top ten tech leaders, among which seven are U.S. firms including Apple Inc. These seven U.S. top tech firms, acting as leading customers in global supply chain, are suitable targets of our study. Meanwhile, we select Taiwan as the local market for the following reasons. Firstly, many enterprises in Taiwan are the major original equipment manufacturers (OEM) of these U.S. firms. For instance, Foxconn Technology Group, making up 6% of the total value in Taiwan stock market, is the largest OEM of Apple Inc.'s products and has more than 50% sales from Apple. Secondly, individual investors dominate the stock market in Taiwan (Hung, Lu, Lee, 2010, Lin, Ko, Feng, Yang, 2016), suggesting that individual investors' limited information access contributes a lot to the equilibrium price (Jarrow, 1980).

Empirically, this study first tests whether there exists return linkages among U.S. tech leaders and their suppliers in Taiwan. Moreover, it's noticed that these suppliers are major members of electronic industry and occupy about half of the industrial market value. Naturally, there is a high possibility that stock returns on U.S. customers not only affect suppliers' returns but also have impacts on the electronic industry index return. Moreover, as electronic industry comprises half of the listed firms in Taiwan stock exchange (TWSE) and takes up over 40% of total market value, market index return may also be affected. These facts drive us to further investigate whether U.S. stock returns also spillover to the electronic industry indices and market index.

If U.S. customers have return spillover effects on the Taiwanese suppliers, electronic industry index and market index, we then cast another reflection on the return diffusion from U.S. stocks to the non-electronic index, deriving from the point of Karolyi and Stulz (1996) that return contagion of different markets occurs when enthusiasm for stocks in one market induce enthusiasm in other markets, regardless of the evolution of market fundamentals. To examine this reflection, we first test whether investor sentiments on the U.S. stocks diffuse to the sentiments on the electronic industry sub-indices and non-electronic industry index. Then, possibly stemming from investor sentiment diffusion, return linkages between U.S. stocks and non-electronic industry index can be investigated.

After verifying the first hypothesis about return spillovers, we examine the second hypothesis by focusing on the sensitivities of individual stocks to each of seven U.S. stock returns. Firstly, sorting on the sensitivity and dividing stocks into ten groups, we observe that portfolios with the highest positive sensitivity to the leading stocks have the highest stock returns. Moreover, this sensitivity significantly and positively affects idiosyncratic volatility. Then, adding a new constructed sensitivity risk factor to the 6-factor (Fama and French, 2015's five factor plus Carhart, 1997's momentum factor) model, we find that significant abnormal returns decrease or become insignificant for the groups with top sensitivities. Besides, the relationship between cross-sectional return and this sensitivity factor is significantly positive.

The contributions of this study are as follows. We contribute to the supply-chain literature by examining stock return spillover effects along international supply chain. Moreover, we point out that stocks with strong positive sensitivity to U.S. stocks, have a return premium and high abnormal returns, providing a new perspective to study asset pricing in local market.

Section snippets

Data

The seven top U.S. high-tech firms are Apple Inc., Amazon, Google, Microsoft, Intel, Cisco Systems and Texas Instruments Incorporated. It is difficult to obtain detailed global supplier information for these listed firms. Since 2012, Apple Inc. annually releases a supplier list containing its last year's 200 top suppliers. We collect the Taiwanese suppliers of Apple from these supplier lists by hand. For other U.S. stocks, Taiwanese suppliers are collected by various resources such as news.

The VAR model

Diebold and Yilmaz (2008); Singh et al. (2010); Natividad et al. (2011) and Huo and Ahmed (2017) use VAR models to investigate return spillovers across markets. Following above studies, we employ VAR models to explore spillovers from each U.S. stock to Taiwan. A lag VAR model (VAR(p)) can be expressed mathematically as where $\mathbf{Y}$ denotes

a $k \times 1$ vector of stationary time series, and λ is the lag value for the corresponding variables, c is a $k \times 1$ vector of

Sensitivity to U.S. firm and idiosyncratic volatility

Similar to Hu et al. (2018), we measure the sensitivity of stock i to each U.S. stock return by the U.S. beta β_{it} from where r_{it} is the Taiwanese stock i 's return on day t , and r_{jt} and r_{kt} represent returns on individual U.S. stock on day t and k respectively. US_{firm} refers to Apple, Amazon, Google, Microsoft, Intel, Cisco Systems or Texas.

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Conclusions

This study aims to explore relationships between return linkages of foreign markets, idiosyncratic volatility and abnormal return in local markets. Empirical results show significant stock return spillover effects from seven top U.S. high-tech firms to suppliers, electronic industry and market index in Taiwan. Besides, we find that returns and investor sentiments on the U.S. stocks also diffuse to the non-electronic index, providing one more evidence on the inadequacy of broadness and depth in

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- [Gradual market development reforms and venture capital strategy adjustments: Evidence based on industrial chain innovation and development](#)

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Citation Excerpt :

Meanwhile, industrial chain innovation and development, as a key to the success of market development reforms, can attract venture capital investments, and through technological progress and business model innovation, optimize and upgrade the economy (Kong et al., 2021). Therefore, studying the relationship between gradual market development reforms and venture capital strategies is crucial for understanding industrial chain innovation and development dynamics (Tsai et al., 2020). Despite extensive literature discussing market development reforms (Goodell et al., 2022; Zhang et al., 2024), venture capital investment (Cumming, 2014; Hsiao and Shiu, 2023), and industrial chain innovation (Xiao et al., 2024; Xu et al., 2023), few studies have systematically analyzed the interaction among these three from the perspective of gradual market development reforms.

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